The Effect of Data Balancing on Performance of Support Vector Machine and Random Forest Classification for Typhonium Flagelliforme Lodd Dataset

Iwan Binanto
Informatics, Sanata Dharma University
Yogyakarta, Indonesia
iwan@usd.ac.id

Bernadetha Mega Devina Ayuningtyas
Informatics, Sanata Dharma University
Yogyakarta, Indonesia
detha00mega@gmail.com

Nesti Fronika Sianipar
Biotechnology Departement,
Faculty of Engineering, Bina Nusantara University
Research Center Food Biotechnology,
Bina Nusantara University,
Jakarta, Indonesia
nsianipar@binus.edu

2024 International Conference on Informatics Electrical and Electronics (ICIEE) | 979-8-3315-1921-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICIEE63403.2024.10920451

***Abstract*—This study investigates the effects of data balancing on Support Vector Machine and Random Forest classification. Unbalanced Typhonium Flagelliforme Lodd plant data were used in this study which balanced by SMOTE, then classifying with Random Forest and Support Vector Machine by sigmoid and RBF kernels on both imbalanced and balanced data. Support Vector Machine struggled to identify the minority class in both scenarios with precision, recall, and F1-score were poor, but accuracy was good. On balanced data, however, Random Forest performed somewhat worse in terms of performance indicators, but still did a great job of identifying the minority class. Meanwhile it had little effect on Support Vector Machine. We conclude that Random Forest performs better when classifying unbalanced data, particularly when identifying minority class despite concerns about overfitting. Meanwhile, Support Vector Machines is not suitable for classification on this dataset.**

***Index Terms*—Imbalanced Data, Support Vector Machine, Random Forest, SMOTE, Typhonium Flagelliforme Lodd plant data**

I. Introduction

Imbalanced data is a significant challenge in machine learning, affecting classification training and potentially leading to biased results towards the majority class [1], [2]. *Typhonium Flagelliforme Lodd* plant data is imbalanced data that has been extensively researched by [3], [4], [5], [6]. Figure 1 illustrates that the majority class has 663228 data points, whilst the minority class has just 9830 data points.

Support Vector Machines (SVM) and Random Forest (RF) are widely used classification algorithms for dealing with imbalanced data, focusing on finding the optimal separating hyperplane and do classifications or predictions [7], although they usually struggled with imbalanced data. Additionally, the SMOTE is a widely used method that generates syntethic data points to balance the dataset, improving model performance on minority classes.

979-8-3315-1921-6/24/$31.00 ©2024 IEEE

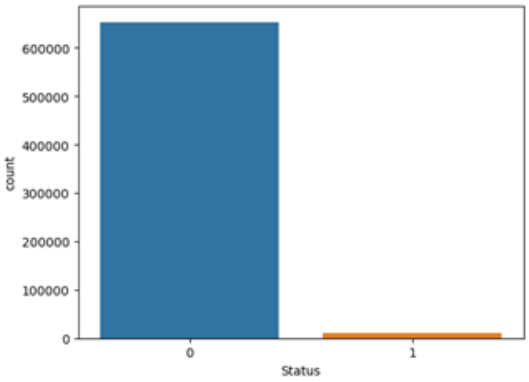


Fig. 1. The Typhonium Flagelliforme Lodd imbalanced dataset.

SMOTE approach is frequently used to deal with imbalanced data in order to have balanced data so that the classification results would be better [8], [9], [10].

II. Literatur Review

Support Vector Machine is a powerful classification algorithm for handling high-dimensional, non-linear data, offering advantages like large feature handling, non-linear decision boundaries, and preventing overfitting [7]. However it has disadvantages, like long computational time and performance can be influenced by parameter settings [11].

Random Forest is an ensemble algorithm that uses decision trees to enhance classification accuracy, efficiently handling large datasets with features, and preventing overfitting [12]. However, it requires long computational time and is less accurate for complex text classification [11].

Authorized licensed use limited to: Badan Riset Dan Inovasi Nasional. Downloaded on August 11,2026 at 17:15:19 UTC from IEEE Xplore. Restrictions apply.

SMOTE is a method used to address imbalanced data by generating synthetic samples from minority classes. It reduces overfitting by increasing the number of samples in the minority class, replicates existing samples, and increases classification accuracy [13], [14], [15]. However, it may be less efficient on datasets with many features and its computing time can be influenced by the algorithm used to find nearest neighbors [13].

III. Methodology

The research method for assessing the impact of data balancing using SMOTE on Support Vector Machine and Random Forest classification for unbalanced data is shown in Figure 2.

The first step is to load the dataset as shown in Figure 3 and then perform preprocessing by deleting the columns "NamaSenyawa" and "RumusSenyawa" as the columns are object-type. After that, the Support Vector Machine methods (RBF and sigmoid kernels) and Random Forest methods are classifying for imbalanced data, then compared to balancing data using the SMOTE method, and then classifying with the same method, and the results are compared and evaluated. The diagram of balanced data is shown in Figure 4

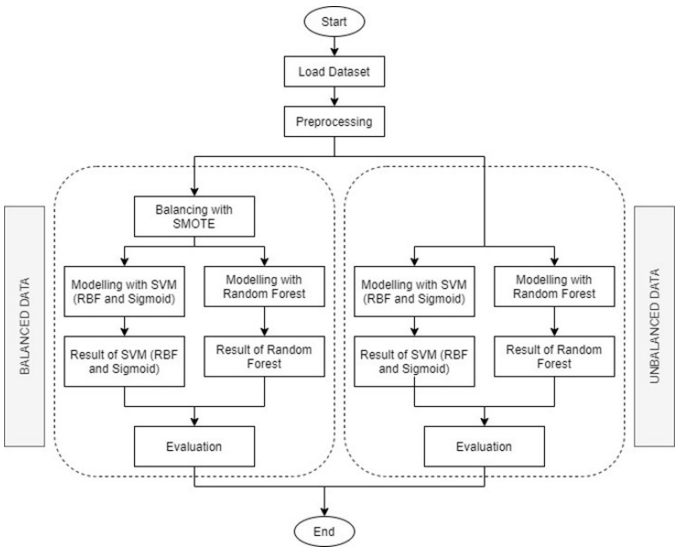


Fig. 2. Research Method

	Retention Time	Intensity	m/z	Real_m/z	NamaSenyawa	RumusSenyawa	Status
0	5.022	281952	60.057205	60.03236	Urea	CH4N2O	0
1	5.022	198032	60.246964	60.03236	Urea	CH4N2O	0
2	5.022	43064	66.888519	67.04220	Pyrrol	C4H5N	0
3	5.022	63172	67.268036	67.04220	Pyrrol	C4H5N	0
4	5.022	57852	68.280075	68.03745	Imidazole	C3H4N2	0
...	...	...	...	...	...	...	...
663223	2401.000	456	1188.677734	1188.42798	Man2Xyl/ManGlcNAcFucGlcNAc	C45H76N2O34	0
663224	2401.000	191	1191.460938	1191.38086	Foetoside C	C58H94O25	0
663225	2401.000	191	1193.358398	1193.40686	Man6GlcNAc-I	C44H75NO36	0
663226	2401.000	191	1195.192871	1195.35303	Furostane base -2H + O-Hex, O-Hex-dHex-dHex-dHex	C57H94O26	0
663227	2401.000	419	1195.572266	1195.35303	Furostane base -2H + O-Hex, O-Hex-dHex-dHex-dHex	C57H94O26	0

663228 rows × 7 columns

Fig. 3. The Typhonium Flagelliforme Lodd complete dataset [16], [17]

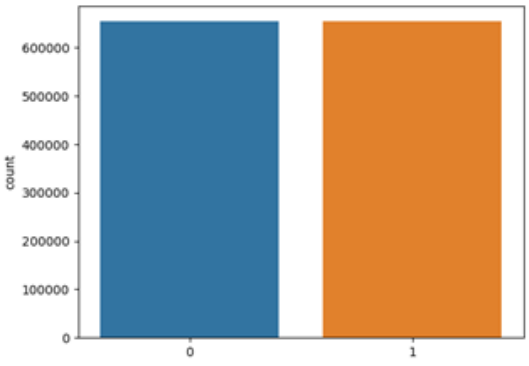


Fig. 4. The Typhonium Flagelliforme Lodd balanced dataset by SMOTE

In Figure 4 default parameters are used for both Support Vector Machine and Random Forest. It is for evaluating the default parameters too, is it good enough or not in this case.

IV. Results and Discussions

Support Vector Machine (SVM) with Sigmoid and RBF kernels perform poorly in identifying minority classes on imbalanced and balanced data. High accuracy doesn't always translate to good model performance, as Precision, Recall, and F1-Score are very low especially on RBF kernel.

TABLE I
Result Of Classification Of Support Vector Machine and Random Forest

	Imbalanced Data			Balanced Data		
	SVM		RF	SVM		RF
	RBF	Sigmoid		RBF	Sigmoid	
Accuracy	0.985	0.970	0.993	0.985	0.970	0.992
Precision	0.0	0.013	0.926	0.0	0.013	0.901
Recall	0.0	0.013	0.607	0.0	0.013	0.580
F1-Score	0.0	0.013	0.733	0.0	0.013	0.706
Time (minutes)	8.456	49.817	3.8958	9.605	49.796	4.888

Support Vector Machine performed poorly or failed in identifying the minority class, achieving high accuracy but very low precision, recall, and F1-score. Random Forest excelled at identifying the minority class, even with a slight decrease in performance metrics on balanced data.

Random Forest (RF), on the other hand, is superior in identifying minority classes, despite a slight decline in Precision, Recall, and F1-Score on balanced data. Random Forest does not perform better when the dataset is balanced using SMOTE. Its accuracy, precision, and recall are lower compared to the imbalanced data. This provides evidence that balancing data using SMOTE in this case is not suitable for Random Forest, let alone Support Vector Machine.

V. Conclusions

The study examined the impact of SMOTE data balancing on the performance of Support Vector Machine and Random Forest for imbalanced data classification. It used imbalanced Typhonium Flagelliforme Lodd plant data. Data balancing

Authorized licensed use limited to: Badan Riset Dan Inovasi Nasional. Downloaded on August 11,2026 at 17:15:19 UTC from IEEE Xplore. Restrictions apply.

using SMOTE significantly improved Random Forest's computational efficiency, but had minimal impact on Support Vector Machine's effectiveness. In conclusions, Random Forest is a more suitable classification algorithm for imbalanced data, especially when identifying the minority class although there are concerns about overfitting. The characteristics of the Typhonium Flagelliforme Lodd data is not suitable for classification using Support Vector Machine especially Sigmoid kernel, because both imbalanced and balanced data yield poor results, including long execution times.

Since Support Vector Machine and Random Forest perform differently with SMOTE, it would be useful to explore other balancing techniques or further tuning of these models.

ACKNOWLEDGMENT

Thank you to Dr. Nesti Fronika Sianipar, S.P., M.Si. for providing the raw data of Typhonium Flagelliforme Lodd to use in this research

REFERENCES

[1] P. Vuttipittayamongkol, E. Elyan, and A. Petrovski, "On the class overlap problem in imbalanced data classification," *Knowledge-Based Systems,* vol. 212, p. 106631, Jan. 2021, doi: 10.1016/j.knosys.2020.106631.

[2] D. Dablain, B. Krawczyk, and N. V. Chawla, "DeepSMOTE: Fusing Deep Learning and SMOTE for Imbalanced Data," *IEEE Trans. Neural Netw. Learning Syst.,* vol. 34, no. 9, pp. 6390–6404, Sep. 2023, doi: 10.1109/TNNLS.2021.3136503.

[3] G. A. Batan, M. J. Keytimu, F. L. Katumbo, I. Binanto, and N. F. Sianipar, "Penerapan Metode Random Forest, Gaussian NB, Dan KNN Terhadap Data Unbalance dan Data Balance Menggunakan Random Over Sampling Untuk Klasifikasi Senyawa Keladi Tikus".

[4] I. Binanto, N. F. Sianipar, F. Dea, M. N. Primadani, and T. W. Kartikasari, "Klasifikasi Senyawa Keladi Tikus Menggunakan Algoritma KNN, Gaussian Naïve Bayes dengan Menerapkan Imbalance Data Borderline SMOTE," *SNST*, vol. 13, no. 1, p. 377, Nov. 2023, doi: 10.36499/psnst.v13i1.9005.

[5] E. Tangkelobo, W. Mayaut, H. Listanto, I. Binanto, and N. F. Sianipar, "Perbandingan Algoritma Klasifikasi Random Forest, Gaussian Naive Bayes, dan K-Nearest untuk Data Tidak Seimbang dan Data yang diseimbangkan dengan metode Random Undersampling pada dataset LCMS Tanaman Keladi Tikus".

[6] A. P. Monika, F. E. P. Risti, I. Binanto, and N. F. Sianipar, "Perbandingan Algoritma Klasifikasi Random Forest, Gaussian Naive Bayes, dan KNearest Neighbor untuk Data Tidak Seimbang dan Data yang diseimbangkan dengan Metode Adaptive Synthetic pada Dataset LCMS Tanaman Keladi Tikus".

[7] Z. Rustam, E. Sudarsono, and D. Sarwinda, "Random-Forest (RF) and Support Vector Machine (Support Vector Machine) Implementation for Analysis of Gene Expression Data in Chronic Kidney Disease (CKD)," *IOP Conf. Ser.: Mater*. Sci. Eng., vol. 546, no. 5, p. 052066, Jun. 2019, doi: 10.1088/1757-899X/546/5/052066.

[8] L. Qadrini, "Handling Unbalanced Data With Smote Adaboost," Mantik, vol. 6, no. 36, pp. 2332-2336, August 2022, doi: 10.35335/mantik.v6i2.2597.

[9] A. Anggrawan, H. Hairani, and C. Satria, "Improving Support Vector Machine Classification Performance on Unbalanced Student Graduation Time Data Using SMOTE," *IJIET*, vol. 13, no. 2, pp. 289–295, 2023, doi: 10.18178/ijiet.2023.13.2.1806.

[10] B. Prasetiyo, Alamsyah, M. A. Muslim, and N. Baroroh, "Evaluation performance recall and F2 score of credit card fraud detection unbalanced dataset using SMOTE oversampling technique," *J. Phys.: Conf. Ser*., vol. 1918, no. 4, p. 042002, Jun. 2021, doi: 10.1088/1742-6596/1918/4/042002.

[11] L. Arnroth and J. Fiddler Dennis, 'Supervised Learning Techniques: A comparison of the Random Forest and the Support Vector Machine', Dissertation, 2016.L. Arnroth and J. Fiddler Dennis, 'Supervised Learning Techniques: A comparison of the Random Forest and the Support Vector Machine', Dissertation, 2016.

[12] D. Ramayanti, "Comparison of Random Forest and Support Vector Machine for Indonesian Tweet Complaint Classification," *IJSRCSEIT*, pp. 202–207, Dec. 2019, doi: 10.32628/CSEIT195628

[13] S. Sreejith, H. Khanna Nehemiah, and A. Kannan, "Clinical data classification using an enhanced SMOTE and chaotic evolutionary feature selection," *Computers in Biology and Medicine*, vol. 126, p. 103991, Nov. 2020, doi: 10.1016/j.compbiomed.2020.103991.

[14] M. Zhang and J. Yang, "SMOTE algorithm applying imbalanced data in higher education," in *Proceedings of the 2nd Information Technology and Mechatronics Engineering Conference (ITOEC 2016TOEC 2016)*, Chongqing, China: Atlantis Press, 2016. doi: 10.2991/itoec-16.2016.33.

[15] R. K. Tripathi, L. Raja, A. Kumar, P. Dadheech, A. Kumar, and M. N. Nachappa, "A Cluster Based Classification for Imbalanced Data Using SMOTE," *IOP Conf. Ser.: Mater. Sci. Eng.*, vol. 1099, no. 1, p. 012080, Mar. 2021, doi: 10.1088/1757-899X/1099/1/012080.

[16] Binanto, I., Warnars, H. L. H. S., Sianipar, N. F., & Budiharto, W. Anticancer Compound Identification Model of Rodent Tuber's Liquid Chromatography-Mass Spectrometry Data. *ICIC Express Letters, 16*(1), pp. 9–16, January 2022, https://doi.org/10.24507/icicel.16.01.9

[17] Binanto, I., Warnars, H. L. H. S., Sianipar, N. F., & Budiharto, W. Webscraping Data Labeling System on Liquid Chromatography-Mass Spectrometry of Rodent Tuber for Efficiency of Supervised Learning Preprocessing. *ICIC Express Letters, Part B: Applications, 13*(1), pp. 107–114, January 2022, https://doi.org/10.24507/icicelb.13.01.107.

Authorized licensed use limited to: Badan Riset Dan Inovasi Nasional. Downloaded on August 11,2026 at 17:15:19 UTC from IEEE Xplore. Restrictions apply.